



Feasibility study of integrating the renewable energy system for increased electricity access: A case study of Choba community in Nigeria



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ABSTRACT

There is no way for developing countries to prosper economically or improve their living standards in rural areas if they lack access to energy and power. Access to electricity is frequently a disadvantage in rural areas of developing nations. This work aims to investigate rural electrification in Port Harcourt, Nigeria, using the Choba community as a case study. The investigation was divided into two stages: the first determined the current state of electrification in Choba, and the second study conducted a detailed assessment of integrating the renewable energy system for increased electricity access using the simulation software HOMER pro to estimate the number of photovoltaic modules (PVMs) and wind turbines required to ensure optimal performance. The analysis was examined by selecting the optimum configuration based on levelized cost of energy (LCOE) and net present cost (NPC). The normal energy consumption for the community is estimated to be approximately 39.12–46.80 MW for full 24-hour supply to all residents. In 2021, 6067.64 megawatt hours of electricity were supplied by Port Harcourt electricity distribution company (PHEDC). From the obtained results, this paper proposes an on-grid hybrid PVM/wind turbine system configuration that yields the optimal results in the case study to solve the power outages currently experienced in Choba. The optimal configuration has an NPC of \$20.8 Million and LCOE \$0.0117/kWh, with an operating cost of \$204.52. This system reduce cost of energy by about 87% when compared with the non-MD customers in Band D.

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Introduction

Global energy demand has seen a steady increase as a result of industrial activity and the impact of emerging technologies [16]. As at year 2020, the global electricity generation has already reached 26,823 GWh, with China being the world's largest producer accounting for almost one third of all (29%), followed by USA, accounting for about 16% [47]. Nigeria is a sub-Saharan African country with the largest economy and a population of over 200 million people [10]. A sizable proportion of the population lives in rural areas with extremely limited access to sustainable energy [46]. This has stymied

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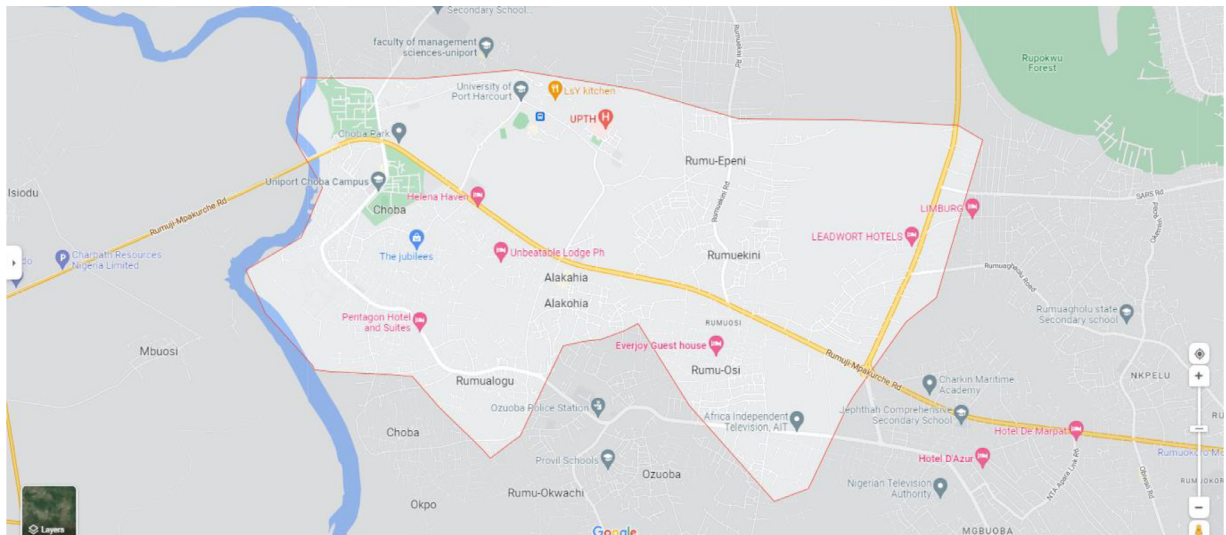


Fig. 1. Location of Choba.

overall development process of the country, as the magnitude of a society's standard of living, as well as the growth and development of its economy is substantially hinged on improvements within the electric power system [31,57]. Without a doubt, the country has been experiencing severe electricity deficit, as the amount of electricity generated is insufficient to power the country 24 h a day [56]. Low electricity generation, combined with the size and population of the country, play an important role which is liable for the low per capita electricity consumption in the country, averaging 184 kWh in 2020 [63]. In 2020, the most common source for electricity production in Nigeria is natural gas (77%), followed by hydroelectric power (20.8%), with solar, bio energy, coal, and other fossil fuels accounting for the remainder [64]. Yet with an installed capacity of about 12.5 GW, only about one-third of this meagre capacity is deployed, which is attribute to a multiplicity of factors, including gas supply constraints, general mismanagement, prolonged underinvestment in infrastructure, weak policy enforcement and a high rate of unpaid electricity bills [15,70]. Nigeria's grid-supplied electricity is grossly inadequate, with about 60% of Nigerians having access to it [23,55]. According to the report of World Bank, as of 2017, 54.4% of Nigerians lacked access to grid electricity, with the figure rising to 56.5% in 2018 and 55.4% in 2020 [67]. Existing electricity companies often have a tradition of serving mostly urban populations. As a result, electricity access is available in 86% of urban regions and 41% of rural areas [28]. Even so, power supply remains a serious issue for those who have access to it, bordering on outages and poor services, as only about 10% of total power demand is met [26]. Worse yet, an estimated 80% of those with access also rely on gasoline or diesel generator sets to deal with power outages, posing serious environmental, health, and climate change concerns [14,70].

Choba, shown in Fig. 1, is a suburb located in the Ikwerre local government area in Rivers State; Nigeria (coordinates 4.8907 N, 6.9034E). It is located on the eastern bank of the New Calabar River, 16 km to the northwest of the Port Harcourt central business district. Choba is a populated place with a total coverage area of about 23.65 km² where people live and work. The community thrives through a variety of small and medium-sized enterprises with bulk of businesses including hairdressers/barbershops, caterers, retail merchants of goods and services such as ice block sales, soft drinks, sachet and bottled water and computer training. From experience, Choba, like the majority of rural villages, has suffered fettered access to electricity, which has had a significant detrimental impact on the livelihood of the community and commerce. The University of Port Harcourt was established in Choba, which means that a significant proportion of its population are students and one of the main drivers of population growth in the region. The lack of electricity at night not only brings utter darkness in the community but has left the community in a state of insecurity [38,66]. This has been made worse by increased pump price of petroleum products used in powering fuel and diesel generators, as well as scarcity, which are allegedly due to their rural location, inaccessible road systems and few retail outlets.

The objectives envisioned by the Federal Government of Nigeria (FGN) are to increase electricity access to 90% by 2025 and source 10% of the total energy supply from renewable energy (RE) sources according to the National Electric Power Policy of 2001 and the Rural Electrification Policy of 2005 [37]. The two main alternatives for this strategy are solar and wind energy for a nation aiming to shift from fossil fuels, particularly gas-based electricity generation model to one that is more dependant on RE sources [13]. In Nigeria, there are plenty of wind and solar resources that can be incorporated into the energy matrix [49]. Nigeria has solar potential of about 4.0–5.0 kWh/m²/day in the south, including the coastal region, 5–5.7 kWh/m²/day in the northwest and north-central region, and 5.7–6.5 kWh/m²/day in the northeast, as well as wind speeds greater than 7 m/s in the North arid zone (Sokoto, Kastina, Kano, Plateau, and Jigawa), and less than 4.4 m/s in the forest and the guinea savannah zones (Abam et al., 2014). As a result, the RE sources of Choba offers the community

a chance to satisfy the electricity demand of the people in a sustainable manner while also advancing the FGN's objectives. Hence, there is a pressing need to amass and judiciously utilize the opportunities offered by these resources. The key to finding a solution is to design a renewable energy system that can make use of the local RE resources.

Recently, electricity tariffs surge by 58% following the removal of power sector subsidies by the Federal Government [7]. With the high electricity cost, the population's relatively low-income levels and exploitation of estimated billing by distribution companies, rural residents are struggling to pay electricity bills. Furthermore, because of lower population densities over larger areas, the cost of providing energy and energy efficiency services to rural households is on average higher than for their urban counterparts. Thus, the main concern in this community is not only to have a flexible and reliable system, but also to have a more affordable electricity system. It is along this thought that this research was embarked on.

One of the most widely used tools for modelling both conventional and RE power systems in the literature is Hybrid Optimization of Multiple Energy Resources (HOMER) [1]. This software application is used for simulation, optimization, and sensitivity analysis for off-grid and grid-tied power systems. HOMER is a dynamic software engine that runs complex simulations of energy data and system components in hybrid electrical systems to determine the least-cost solution and most effective risk-mitigation strategies. It delivers an accurate, versatile, and reliable techno-economic simulation [9], and is simple to use for modelling and designing RE systems [61]. Its library has a large selection of technologies and different components that make it useful for simulating energy systems. Additionally, [11] justify that HOMER outperforms other tools when it comes to sizing RE power systems. In this regard, this tool has been extensively used for rural electrification in places like Ijebu-Itele, Ogun State [53], Lade II, Kwara State [33], Giri, Abuja [59], Irele, Ondo State [18] and Umuejim Amorka, Anambra State [54]. Therefore, this software was a good choice for this investigation.

Wind and solar energy have risen to prominence in recent years because of their significant impact on the world's installed capacity [32,36] and the low levelized cost of energy (LCOE) [58]. Therefore, a handful of enterprising studies are now exploring solar and wind RE using HOMER software. [48] presented a case study for the application of photovoltaic as an energy enabling technology for rural electrification in India. For the remote locale, the PV/battery hybrid system proved a realistic and cost-effective solution. [8] designed and analysed a grid-connected PV under different tracking systems using Makkah as the study location. They evaluated the effects of different tracking designs on the power generated by PV panels. The results showed that the vertical axis tracker with continuous adjustment is the best solution since it has a low LCOE and NPC value, a positive ROI and a high RE penetration to the grid. In literature, a few studies have examined the feasibility of deploying the off-grid or grid-connected systems for electricity production in Nigeria [2,6,45]. [2] examined the feasibility of solar PV-grid tied energy system in supplying electricity to the study location Jos, in Plateau state. According to the study, the energy system can create 331,536 kWh of electricity per year, with solar PV contributing 40.4%, and the LCOE is \$0.103/kWh. On the other hand, [51] evaluated four vertical-axis wind turbines (EOLO-3 kW, Spiral-1 kW, Turby-2.5 kW, WRE-3 kW) and a tree shaped wind turbine (Aeroleaf-11.7 kW) in 8 selected stations in South Africa. The results showed that when using the EOLO wind turbine, the Port Elizabeth station had the lowest LCOE of -0.363 \$/kWh, while the Bloemfontein station had the highest LCOE of 1.601 \$/kWh. [25] carried out a study that estimates the wind power generation capacity of northern and southern Oman and discusses the selection of the most economical, efficient and reliable wind turbines in Oman. The DW54 turbine, with a COE of \$0.119/kW, was recommended for generating electricity in northern Oman, while the Hummer H25.0–200 kW turbine was recommended for south Oman, with a COE of \$0.070/kW. Several studies have been carried out to assess the technical and feasibility analysis of various hybrid PV/Wind energy systems. [72] studied the technical feasibility of a hybrid PV/Wind system to supply electric power to a medium-size workshop in an industrial area in Ardabil, Iran. The results showed that the net present cost was \$304,380 and LCOE is 0.462 \$/kWh. [13] assessed solar-wind power on-grid system systems in the Caribbean region of Colombia using HOMER simulation tool. In Nigeria, [30] evaluated a wind solar hybrid energy system for an electric vehicle charging station. They came to the conclusion that the hybrid system could meet 44.4% of the yearly electricity needs with wind energy and 55.6% with solar energy for an LCOE of 0.064 \$/kWh. According to these studies, increasing the capacity of solar and/or wind energy systems might be used to ease the nation's energy crisis and lower greenhouse gas emissions.

The key problem in using HOMER is to generate a realistic load profile in order to provide practical optimal designs. An accurate load prediction is necessary for effective energy planning [27,68] assessed the techno-economic impact of a distribution generation system with solar PV energy for coastline rural communities in the Niger Delta region of Nigeria. The information utilized to develop the load profile was acquired through surveys, personal interviews with residents and phone interviews. Besides the fact that phone interview are unable to capture body language of the respondents and can limit the complexity of questions [40], the majority of these rural dwellers cannot afford mobile phones and have no phone service [29]. However, because the interview may create bias, the method of data collecting used in the study cannot guarantee realistic results. In [73], the authors performed the techno-economic feasibility of 3 energy sources comprising of hydropower, solar and biogas energy sources for an ICT building in the Federal University of Technology, Nigeria. A drawback of this study is the lack of exhaustive load data collection and analysis. A lack of vital data is one of the key causes of the current difficulties in load prediction. It is more reasonable to use real data so as to allow prospective investors to identify important risk and uncertainty factors and forecast future trends.

Therefore, the perspective examines the issue of Choba electrification and offers a long-term solution. To date, no detailed analysis of the techno-economic feasibility of renewable energy system for Choba has been conducted. Accordingly, the main objective of this research is to highlight the renewable energy potential of Choba community of Rivers State, demonstrate its potential to fulfil the rising energy demand, and advocate for renewable energy as a perfect solution to its challenges.

Furthermore, the load profile estimated for this study analysis were based on real data gathered from the distribution feeder (11 kV), via the data logger. The results and analysis from this study are more factual and could encourage the government, investors and key stakeholders in the electricity industry to support a nationwide transition to clean energy systems [5]. Also, the community will be better informed about the RE options accessible to it and how it may contribute to addressing its rising energy needs. This study also validates the benefits of employing RE sources to meet energy needs in a practical and affordable way. Additionally, there is need to supplement the existing information on these topics, and enough research has not been conducted in this subject matter particularly in this location.

Overview of the Nigerian electricity sector

A brief history of the Nigerian electricity sector

The electricity sector value chain in Nigeria was entirely controlled by a state-owned organization- National Electricity Power Authority (NEPA) from 1972 to 2005. However, a liberal government was elected for two terms (1999–2003 and 2003–2007) and the FGN's policies started to have a more liberal economic profile, which resulted in NEPA being unbundled and renamed Power Holding Company of Nigeria (PHCN) in 2005. This resulted in the formation of three distinct entities: six independent generating companies (GENCOs), the Transmission Company of Nigeria (TCN) and eleven independent electricity distribution companies (DISCOs). The Nigerian Electricity Regulatory Commission (NERC) was also created as an independent regulator for the sector. Thus, the subsequent administration (2010–2015) implemented a new sectoral reform with a liberal profile in 2013 as part of a larger plan to reform the industry and spur growth. The eleven DISCOs and six GENCOs were privatized under the new regulatory frameworks while maintaining full ownership of the Transmission Company of Nigeria (TCN). Even now, the Nigerian electricity sector is still deteriorating despite these reforms. The eleven electricity distributions have been licensed to supply electricity to customers in their respective service areas (Figure SM1).

Electricity generation

The electrical power system in Nigeria is pivoted on four basic energy generation technologies: on-grid, embedded, off-grid and captive generation. On-grid electricity generation is a situation in which electricity is generated and connected to the high voltage national grid or utility grid for despatch. Sending excess power generated to the utility grid is a feature of this system. However, in the event of a grid outage, this type of system will be unable to generate electricity. In the studies of (Adaramola, 2014) and [45] the feasibility and economic analyses of an on-grid system using HOMER software were investigated. Off-grid electricity generation is a standalone power system which is highly popular for small clusters such as rural communities, industrial clusters and housing estates amongst others, providing them electricity through independent electricity distribution network systems. However, surplus energy production may go to waste unless one has a low-cost, easy-to-maintain method of storing excess generation, as battery storage is prohibitively expensive, difficult to maintain, and requires a significant investment of time and technical know-how. The authors [6,55] and [19] used the HOMER software for energy designs and techno-economic analysis of off-grid hybrid systems. Embedded generation is electricity generation directly connected to and evacuated through a distribution network. Embedded generation opens up significant opportunities for private sector engagement, while also creating a well-regulated and transparent market for all generators and satisfying energy demand with little environmental impact. Captive generation implies that electricity is entirely consumed by the generator and not sold to a third party, with no transmission or distribution losses or even theft of electricity. Except for "captive generation," which only requires a permit from the NERC, all generators require licenses to operate.

Power transmission network in Port Harcourt

Port Harcourt region currently controls six transmission substations: Ahaoda, Port Harcourt Mains, Port Harcourt Town, Yenegoa, Rumuosi, and Elelenwo (Fig. 2). Rumuosi transmission station receives its energy from Port Harcourt mains sub transmission station and Omoku gas turbine at 132 kV. Rumuosi transmission station is equipped with two transformers rated at 40 MVA, 132/33 kV (T1) and 60 MVA, 132/33 kV (T2), as well as four load feeders: UPTH, NTA, Rukpokwu and New Airport Road, each of which has a maximum load capacity of 15.6 MW, 13.4 MW, 10.8 MW and 0.1 MW respectively. The UPTH lines serve the Uniport and Choba injection substations as well as UPTH substation. Choba injection substations feed Aluu, Choba, and Rumuekini feeders, which in turn serve various parts of these communities.

Nigeria electricity prices

The cost of electricity is fixed and regulated by the NERC [12] and a new service-based tariff went into effect on February 1, 2022 [34,52]. Customers are now divided into bands based on how many hours they use electricity every day. Maximum demand (MD) customers each have their own transformer. In contrast, non-MD customers share transformers with communities. Non-MD customers have single or three-phase connections, whereas MD1 customers have low voltage maximum demand connections and MD2 customers have medium/high voltage maximum demand connections. The price is different for each of the eleven electricity distribution companies. Figure SM2 shows the electricity tariffs of the Port Harcourt Electric Distribution Company, which serves the states of Rivers, Bayelsa, Cross River and Akwa Ibom.

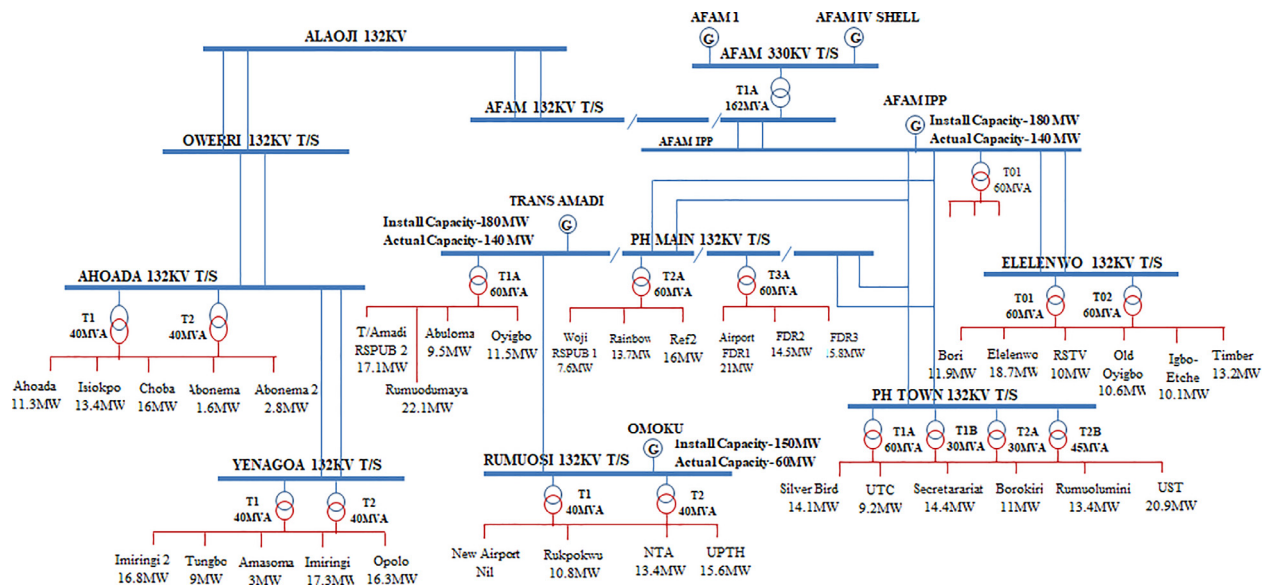


Fig. 2. Diagram of Port Harcourt work centre stations and their load capacity.

Materials and methodology

Data acquisition

The data source for this research work are based on visits made to the injection substation in the community. The hourly electricity demand of the case study community was obtained for 2021. The accuracy of the results of this study is entirely dependant on the accuracy of this information.

Data analysis (GIS study)

The map of Choba was created in ArcGIS environment. The buildings were digitized and the river was extracted from the satellite image of the area using normalized difference water index (NDWI). The wind direction data was acquired from National Aeronautics and Space Administration (NASA). The wind direction data was saved in Excel and imported into ArcGIS environment and was converted into a point data. In addition, digital elevation model (DEM) data was acquired from the USGS earth explorer website and used to calculate the area solar radiation. The DEM data was imported into ArcGIS environment and the solar radiation for the area was determined.

Results and discussion

GIS result

The map of Choba in Figure SM3 displays the distribution of buildings, the direction of the wind, and the electricity transmission network. The wind direction in Choba is 216°, which points toward the south-west. According to the study, Choba has approximately 3261 buildings. The solar radiation distribution of the community is shown in Figure SM4. The map shows that the solar radiation distribution ranges from 4400 Wh/m² to 4500 Wh/m².

Current state of electrification in Choba

Choba is currently powered by the national grid, with private electric generators and uninterrupted power supply devices serving as a backup power supply system for residential homes, shops and offices in the event of power outages or loss of grid connection. In Nigeria, distribution companies categorize the feeders within their franchise areas into service bands. Choba is served by a feeder that belongs to service band D. This implies that electricity consumers are expected to receive minimum of 8 h and a maximum of 12 h of electricity daily. Fig. 3 illustrates the monthly electricity consumption in Choba for 2021. In times when there is a fault, frequent blackouts can stretch for hours, sometimes days, leaving homes and businesses without power. The current situation in Choba demonstrates that its residents face blackouts lasting two to five days. It also reveals that the amount of time spent in periods of light is below the required amount of time allocated to

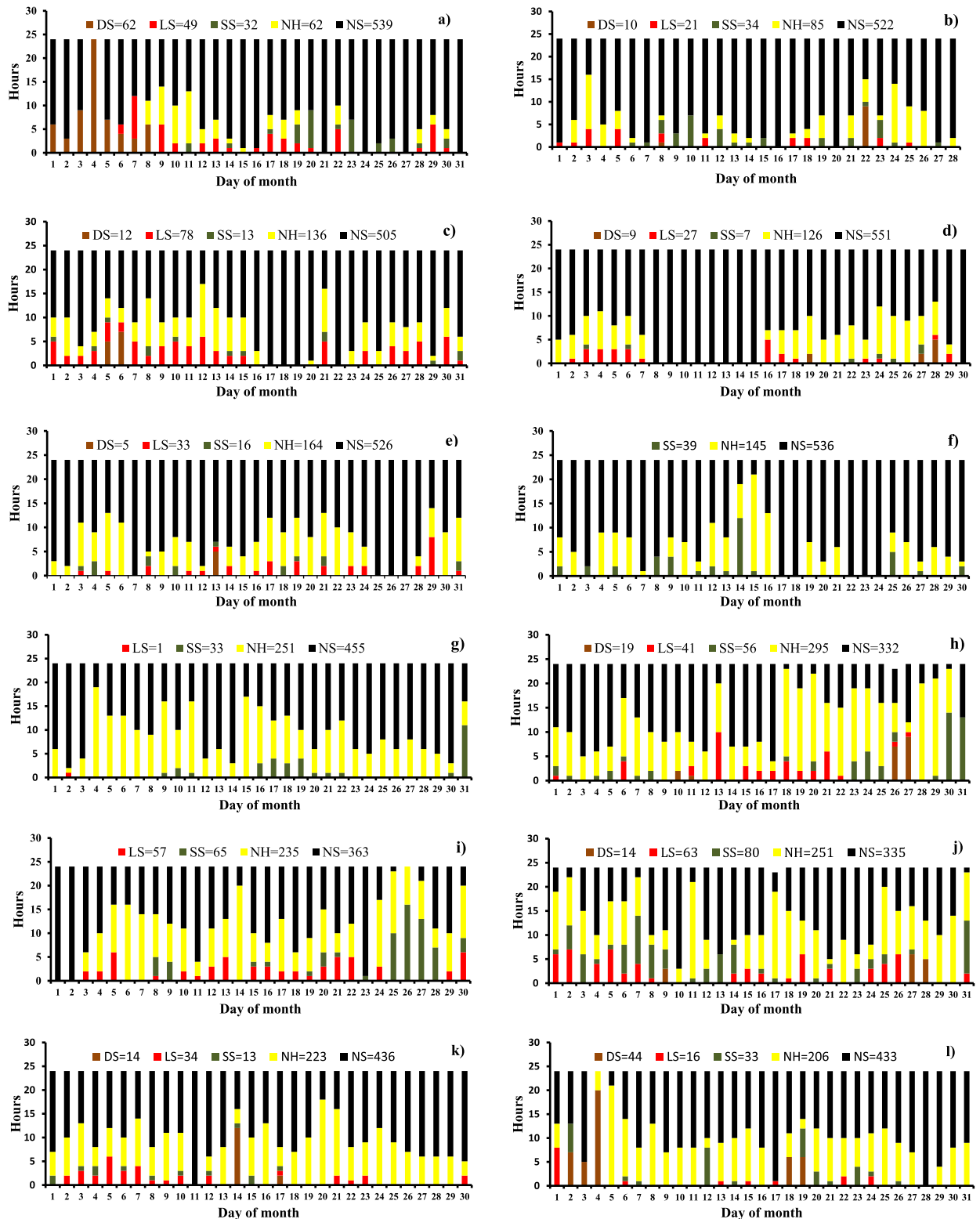


Fig. 3. Electricity supply to Choba for the month of January (a) to December (m) 2021 (NH= net number of hours of electricity supply, DS= disconnection, LS= low electricity supply, SS= service shutdown).

the community. The first half of the year experienced poor electricity compared to the other half. The low supply (LS) is often due to overloading on the network. The afternoon is a time of high demand, so during this time, the voltage tends to decrease naturally and causes lights to become dim. Although LS are not total power failures, they can overheat and damage electrical equipment. The long disconnection hours (DS) are as a result of damage to the electric transmission lines, substations or other parts of the distribution system, as well as faults at the power station. The shutdown services (SS) are used for executing clean-up task and maintenance servicing of electrical installations on the system. Figure SM5 shows the hourly electricity supplied per month for year 2021. In January, February, June, July and November, December 2021, the community is seen to have 62, 85, 145, 251, 223 and 206 h of electricity, respectively. The months with the lowest and highest number of hours are January (56) and August (295), respectively. Choba received light for a total of 181.58 hrs/yr, which is less than the monthly average of 240 hrs/yr. Fig. 4 shows the hourly electricity load for year 2021. The histogram shows a random distribution with no apparent pattern. The normal average energy consumption is estimated to be approximately 39.12–46.8 MW for full 24-hour supply to all residents. In 2021, 6067.64 MWh of electricity were consumed by the community (Figure SM6). During the time period under consideration, there was no discernible trend in the community's energy supply.

Problems associated with the current electrification method

According to the findings of this study, the following underlying problems associated with the current electrification method include:

- The instability of electricity supply from the national grid due to low generation, coupled with a deteriorating an ageing infrastructure within the electricity network. For example, Nigeria's power grid collapsed on March 14, 2022, dropping from 4174.7 MW at 10 a.m. to 199.18 MW around 12 p.m. This was due to the simultaneous shutdown of three power plants totalling 1000 MW: Afam Six, Calabar Plant, and Agip Okpai. The Afam Six plant needed maintenance, and the Calabar facility had problems with gas supply and pipelines. Agip Okpai was vandalized despite producing only 236 MW of its maximum 375 MW capacity [54].
- High number of illicit connections of transmission lines by unscrupulous individuals which are one of the major reasons.
- Non-issuance of bills to some community users, non-reading of meters, use of inappropriate billing methods are experienced at the distribution system.
- Use of load limiters and power theft are experienced at the distribution system.
- Vandalism on PHEDC assets by hoodlums: As an example, the vandalism of a 300 KVA, 11/415 V transformer of the Choba substation in February 2022, resulted in an indefinite outage for hundreds of consumers [69].

Long-term solution to the electricity problem

Embarking on providing power to rural communities is a huge difficulty. Expanding the national grid is expensive and Nigeria consistently lacks the capital to grow their current infrastructure. Other alternatives can be harnessed such as renewable energy. Many countries have transitioned away from gas and toward renewable energy. The need to reduce the carbon footprint as a means of addressing climate change has made the adoption of renewable energy more significant.

Wind is used to generate electricity by utilizing the kinetic energy created by moving air. Wind turbines are used to capture this wind and convert it to electricity. Wind power, by its very nature, is a superior source of renewable energy because it is free, abundant, cost-effective and environmentally friendly [21]. It is more cost efficient than solar energy per unit of electricity output, and it's more sustainable than hydro sources and has the lowest relative greenhouse gas emissions [35]. As of the end of 2020, the cumulative installed wind energy capacity on a global scale totals 733 GW [47,50]. Nigeria has a great potential for electricity generation from wind energy. As part of attempts to develop wind power project in Nigeria, a 10 MW wind farm has been established at Lambar Rimi district of Katsina, Nigeria. The facility consists of 37 wind turbines with a rated power of 275 kW each. The project, which has not yet been put into operation, is the nation's first wind-based energy development [17]. However, wind projects are best suited in Nigeria's northern states, which have the greatest wind energy potential, according to the wind speed chart in Figure SM7 [42]. A well-chosen wind turbine, on the other hand, might be employed in regions where there is little wind.

Solar energy, the most abundant energy source on Earth, is the fastest-growing renewable energy sector with recent annual growth rates of about 23% in 2020 [43]. Solar energy can be converted directly into electricity by photovoltaic cells. Solar power is an infinite source of energy and, as a cost-competitive technology that is constantly growing, it plays a critical role in ensuring energy availability in rural regions. No greenhouse gas emissions are discharged into the atmosphere when solar photovoltaics are utilized to generate electricity. Solar panels have relatively low operating costs when compared to other forms of energy generation [60]. Solar photovoltaics are easier to construct, but they take a great deal of area, and the more electricity you want to generate, the more solar panels you need. China has 208 GW of photovoltaic capacity, which is about 30% of the world's total capacity in 2020 [62]. This makes China the biggest producer of solar energy in the world [22]. Nigeria is currently building a 200 MW solar power facility in Delta state, which is being promoted by B&S Power Holding PTE. LTD, a Singaporean renewable energy company, and Sunnfyred Global, a Nigerian investment entity [39].

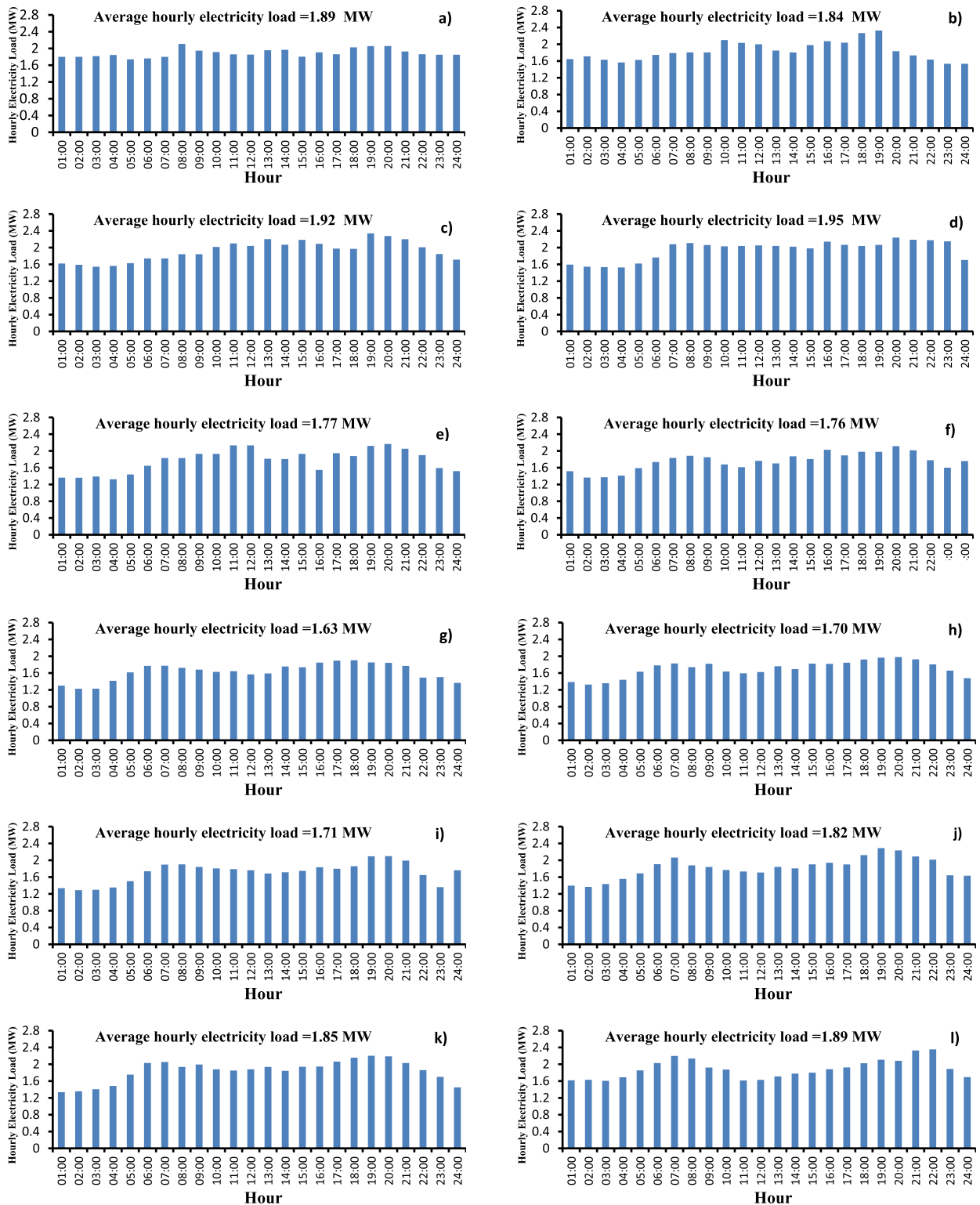


Fig. 4. Monthly mean electricity load distribution for the month of January (a) to December (l) 2021.

Imports of solar devices totalled \$150 million in 2019, and the market for solar home systems and mini-grids in Nigeria is reportedly worth \$9 billion annually [4].

Simulation using HOMER software

Energy system modelling

For this study, we construct a system that can supply the community with flexible, affordable, and reliable electricity. The community's load profile is based on the hourly electricity demand profiles from 2021. The peak and average consumption of the houses are 3823.8 kW and 43,460 kWh/day, respectively, with a load factor of 0.47. The load data is fed into the HOMER software, which generates a graphical representation of the daily, seasonal and yearly load profiles, as shown in Figure SM8. Figure SM9 depicts the suggested system configuration for simulations for this study. The energy generation technology used was a grid-connected power system. As a result, when production exceeds demand, extra energy is fed back into the grid and when renewable resources are not available, electricity from the grid provides the community with the necessary electricity. In accordance with the architecture, there are four main components to the system; PVM, wind turbine, converter and the grid. The AC and DC lines are connected via the converter; the DC power from the PVM is converted into AC power and fed into the grid system using the converters.

Energy resources

The data for this study was obtained from the National Aeronautics and Space Administration (NASA) via HOMER utilizing the location's latitude and longitude. The monthly wind speed profile within the study location is depicted in Figure SM10. At the anemometer height of 10 m, it was discovered that the annual average wind speed available was 3.15 m/s. The minimum monthly wind speed occurs in October (2.35 m/s) while the maximum occurs in March (3.30 m/s). Figure SM11 depicts the generated monthly average global horizon irradiance of the community. The solar irradiation ranges between 3.465 kWh/m²/day and 4.901 kWh/m²/day, whereas the annual average solar irradiation for this region is 4.401 kWh/m²/day. From November to April, the global horizon irradiance rises above the average, with a peak in February. Figure SM12 shows the average monthly temperature of selected area under study. The annual average temperature for the considered location is 22.59 °C, according to the temperature profile. The months with the lowest and highest average temperatures are August (24.52 °C) and March (26.73 °C), respectively.

System component specifications

Grid

The price of electricity for residential households for service band D consumers in May 2022 is 0.0934 \$/kWh, while it costs 0.130 \$/kWh for government agencies, public street lights, commercial and industrial customers (see Figure SM2). Accordingly, this study adopts an average grid electricity cost of 0.0934 \$/kWh (about N38.81/kWh) in its analysis. The excess electricity generated will be sold back to the grid at 10% discount. The electricity rate in Nigeria is anticipated to rise over the life of the system due to the projected elimination of electricity subsidies. As a result, the price of electricity has been updated to reflect these adjustments to be between 0.00934 \$/kWh and 0.1134 \$/kWh. For the purposes of this study, the currency rate used in this study is 1 \$ = N415.63 on June 6, 2022 [24].

Photovoltaic module (PVM)

The photovoltaic module used in this study is a 104-cell (8 × 13) mono-crystalline (model number SPR-MAX3-400, manufactured by Sunpower) with a rated power of 400 W and can produce a maximum 1000 V DC. Sun power panels take the top spot for the most efficient on the commercial market. The detailed technical specifications of this model are presented in Table 1. The lifespan of the panel is given to be 25 years. The capital cost of the PVM is specified as \$1072/kW (<https://www.europe-solarstore.com/sunpower-spr-max3-400.html>). The PVM replacement is assumed to be \$1072/kW with

Table 1
Technical specification of PVM [20,65].

Parameter	Specification
Cells	8 × 13 (104)
Maximum power output	400 W
Open circuit voltage	75.6 V
Temperature	−40 °C to +85 °C
Temperature Coefficient	−0.29
Module efficiency	22.6%
Panel Dimensions:	1690 × 1046 × 40 mm
Power tolerance	±5%

Table 2
Technical specifications of wind turbine.

Parameter	Specification
Start-up Wind Speed	3.4 m/s
Cut-in wind speed	2 m/s
Rated wind speed	12 m/s
Cut-out wind speed	34 m/s
Rotor Diameter	82 m
Rotor speed, max	18.0 U/min
Hub height	85 m
Generator type	Synchronous multi-pole

Table 3
The search space optimization values.

Component	Search space
PVM (kW)	0–17,000
Wind turbine number (kW)	0–15
System converter (kW)	0–9000

no maintenance cost. The derating factor and the ground reflectance of 90% and 20%, respectively was considered for the study.

Wind turbine

The wind turbine used for this study is an Enercon E-82 E2 with a 3-bladed upwind turbine and a capacity of 2 MW. The wind turbine is engineered for excellent reliability, little maintenance and ideal for sites with medium wind speeds. The grid-connected configuration will be used for this work. It has a rotor diameter and hub height of 7 m and 24.3 m respectively. Other specifications are shown in Table 2. In this project, wind turbines were specified with installed capital cost of \$2 million. The replacement cost of the wind turbine considered in this case is \$2 million after 20 year service life. For economic analysis, 1.5% of the replacement cost is assumed for the maintenance cost.

Converter

A converter is necessary for any system that includes both AC and DC components. The total initial, replacement, and maintenance costs of the converter are assumed to be \$550/kW, \$550/kW, and \$10/year, respectively [44]. An efficiency of 95% and 15 years operational lifetime is considered for this study.

Search space

Table 3 displays the input for each component involved in this study. They include the size of the PVM, the number of wind turbines, and the size of the system converter.

Economic analysis

Net present cost (NPC)

The net present cost (NPC) is used to calculate the present value of all expenditures associated with installing and operating the component over a project's lifetime, minus the present value of all revenues earned during that time. NPC is a dependable predictor of the worth of a project, and it is commonly used in economic engineering to assess feasibility [41]. Eq. (3) is utilized to determine NPC [25].

$$NPC = \frac{C_{ann, total}}{CRF(i, N)} \quad (1)$$

where

$C_{ann, tot}$ = Annual total cost

N = project life span

i = discount rate

CRF = Capital recovery factor: which is the capital recovery factor is a ratio used to calculate the present value of an annuity. The equation for the capital recovery factor is:

$$CRF(i, N) = \frac{i(1+i)^N}{(1+i)^N - 1} \quad (2)$$

$$i = \frac{i_o - f}{1 + f} \quad (3)$$

where

i = real interest rate
 i_o = nominal interest rate
 f = annual inflation rate

The project lifetime has been considered to be 25 years. The current nominal interest rate and inflation rate in Nigeria are 11.5% and 15.7% respectively (<https://tradingeconomics.com/nigeria/interest-rate>).

Levelized cost of energy (LCOE)

A key indicator used in the evaluation of energy projects is the LCOE. The LCOE of a project is a ratio of the net present value of the total cost incurred in completing and operating the project to the total electrical energy generated over its lifetime. The LCOE equation is presented in Eq. (6) [71].

$$LCOE = \frac{C_{ann, total}}{R_{prim} + R_{tot, grid sales}} \quad (4)$$

where R_{prim} is the primary load (kWh/year) and $R_{tot, grid sales}$ is the total grid sales (kWh/year).

Renewable fraction (RF)

The renewable fraction (RF) is the total amount of power generated by the renewable energy sources compared to total power generation from the overall system. This can be obtained using Eq. (8) [19]:

$$RF = \left(1 - \frac{\sum E_{nonren}}{\sum E_{served}} \right) \times 100 \quad (5)$$

where E_{nonren} denotes the non - renewable electrical production (kWh/yr), and E_{served} denotes the total electrical load served.

Homer calculations

Photovoltaic modules (PVM)

The output power (P_{PV}) generated by the PVM can be calculated from the following equation [3]:

$$P_{PV} = P_{Rc} f_{PV} \left(\frac{G_T}{G_o} \right) [1 + \beta (T_c - T_o)] \quad (6)$$

where P_{Rc} is the rated capacity of the PV panel under standard test conditions in kW, f_{PV} is the PV derating factor [%], G_T is the solar radiation incident on the PV array in the current time step [kW/m²], G_o is the incident radiation under standard test conditions [1 kW/m²], β is the temperature coefficient of power [%/ °C], T_c is the PV cell temperature in the current time step [°C] and T_o is the PV cell temperature under standard test conditions [25 °C].

Wind turbines

The wind speed at the hub height of the wind turbine is calculated using Eq. (7) (the logarithmic law) [25]:

$$U_{hub} = U_{anem} \frac{\ln\left(\frac{z_{hub}}{z_o}\right)}{\ln\left(\frac{z_{anem}}{z_o}\right)} \quad (7)$$

where U_{hub} is the wind speed at the hub height of the wind turbine [m/s], U_{anem} is the wind speed at the hub height of the anemometer [m/s], z_{hub} is the hub height of the wind turbine [m], z_{anem} is the anemometer height [m], z_o is the surface roughness length [m]. In a case the power law is chosen, wind speed at the hub height is calculated using the following equation, α is the power law exponent:

$$U_{hub} = U_{anem} \left(\frac{z_{hub}}{z_{anem}} \right)^\alpha \quad (8)$$

The expected power output from the hub height wind speed under standard conditions of temperature and pressure is derived from the wind turbine's power curve shown in Figure SM13.

Essentially, the output power of the wind turbine is calculated using Eq. (9):

$$P_{WT} = \left(\frac{\rho_a}{\rho_o} \right) P_o \quad (9)$$

where ρ_a denotes the actual density of air [kg/m³], ρ_o denotes the air density at standard temperature and pressure and P_o denotes the wind turbine power output at standard temperature and pressure [kW].

Also, the output power of the wind turbine (P_{WT}) at a particular site is given as [45]:

$$P_{WT} = \frac{1}{2} \rho_a A V^3 C_p \quad (10)$$

where ρ_a = density of air (kg/m³), A is the swept area of the rotor blades (m²) and V is the wind speed (m/s), C_p = coefficient of power.

Results and discussion

Optimization results of the proposed systems

The results and discussion of the three system designs are presented in this section. The categorized optimization results for the grid connected systems is displayed in Table 4. The optimization results for the hybrid system (category A) clearly revealed the most feasible optimized configuration with a PVM of 7500 kW capacity, a converter of 5000 kW and 10,000 kW of wind generation capacity. The architecture of category B system was composed of 14,500 kW PVM and 9000 kW converter while category C consists of 10,000 kW of wind generation capacity only. The three categories have comparable initial investment costs and provides good payback of about 9 years. For category A, the results clearly reveals that the total yearly electricity produced by the energy system is 26,588,269 kWh, with 10,437,521 kWh coming from PVM, 9477,742 kWh from wind turbine and the remainder 6673,007 kWh from grid purchases. This translates to a PVM capacity factor of 15.9% and a wind turbine capacity factor of 10.8%. The yearly energy consumption is calculated to be 25,772,289 kWh (61.6% is consumed by the community, 38.4% is sold back to the grid). The NPC for category A is \$20.9 Million. This is comparable to categories B and C, but the operation and maintenance costs are relatively high, bringing the operating costs to \$50,768/year and \$52,742/year, respectively. Consequently, category A is the optimal energy system. The system produce energy at a cost of \$0.0117/kWhr which is 87% less than the current Band D non-MD electricity tariff in Choba. The investment has a 9.5-year discounted payback period and a 3.8-percent internal rate of return.

Table 4
Optimization analysis of the energy mix system.

Parameters	Category A	Category B	Category C
<i>Component</i>			
PVM (kW)	7500	14,500	–
Converter (kW)	5000	9000	–
Wind turbine	5	0	10
<i>Cost</i>			
Initial capital cost (\$)	20.8 Million	20.5 Million	20 Million
Total O&M cost (\$)	272,862	1.2 Million	8.3 Million
NPC (\$)	20.8 Million	24 Million	23.6 Million
Operating cost (\$)	204.52	50,768	52,742
LCOE (\$/kWh)	0.0117	0.0127	0.0128
<i>Economics</i>			
Return on investment (%)	2.3	2.4	1.8
Internal rate of return (%)	3.8	4.4	2.9
Simple payback period (yrs.)	13.98	13.66	14.67
Discounted payback period (yrs.)	9.54	9.38	9.87
<i>Electricity production and consumption</i>			
PVM supply (kWh/yr)	10,437,614 (39.2%)	20,179,207 (69%)	Nil
Wind turbine supply (kWh/yr)	9499,809 (35.7%)	Nil	18,955,483 (70.9%)
Grid purchases (kWh/yr)	6668,249 (25.1%)	9054,630 (31%)	7793,756 (29.1%)
Total energy produced (kWh/yr)	26,605,673 (100%)	29,233,837 (100%)	26,749,239 (100%)
AC primary load	15,862,900 (61.6%)	15,862,900 (58.1%)	59.3
Grid Sales	9909,389 (38.4%)	11,420,249 (41.9%)	40.7
Total energy consumed (kWh/yr)	25,772,289 (100%)	27,283,149 (100%)	100%
Excess energy (kWh/yr)	327,898 (1.23%)	991,292 (3.39%)	0%
Renewable fraction (%)	74.1	66.8	70.9

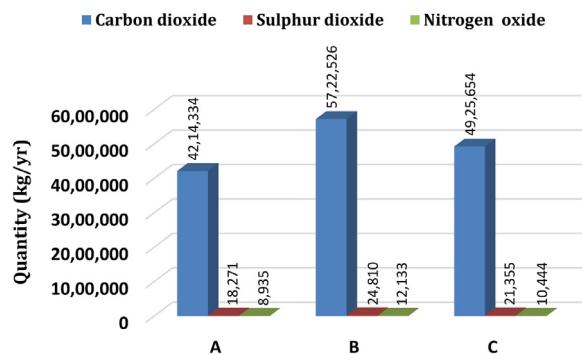


Fig. 5. Total emissions produced by renewable energy sources for the selected systems.

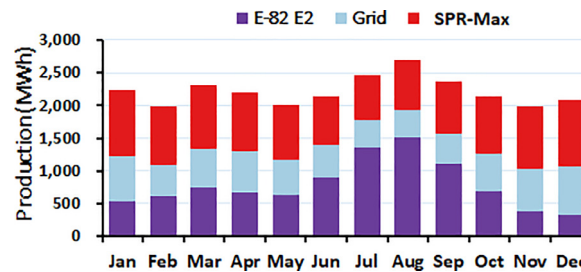


Fig. 6. Monthly distribution of the electricity produced by the energy system.

Table 5

Monthly amount of electricity purchased from and sold to the grid.

Month	Energy Purchased (kWh)	Energy sold (kWh)	Peak Load (kW)
January	681,279	731,042	3424
February	477,017	701,133	2847
March	594,249	761,373	3142
April	629,018	723,454	3522
May	535,583	654,755	3160
June	496,364	815,607	2727
July	421,310	1163,938	2852
August	414,595	1376,082	2668
September	451,785	1064,886	3044
October	571,815	743,900	3129
November	655,403	592,016	3536
December	739,831	581,204	3578
Annual	6668,250	9909,389	3578

Emissions produced in kg/yr for the three cases are shown in Fig. 5. Category A produced the least emissions when it comes to nitric, sulphur, and carbon dioxide which is 8935, 18,271, and 4214,334 kg/yr, respectively.

Assessment of energy production, consumption and cost of the optimal system

The monthly and seasonal variations in amount of electricity produced by PVM, wind turbine and contribution from the grid can be observed from Fig. 6. The monthly distribution of electrical generation shows that the power generation from PVM was higher during the dry season which typical runs from October to April than the rainy season. High temperatures and low humidity characterize the dry season, which is influenced by warm winds flowing in from the Sahara Desert over West Africa. The monthly distribution of electrical generation from the wind turbine shows that the high-wind generation (July-Sept) coincides with the raining season and August fetches the highest generation in wind energy. The South Western wind is a result of the dominance of the tropical maritime air mass from the South Atlantic which causes the rainy season. Table 5 shows the monthly energy purchases and sales to the grid. The annual energy purchased from the grid is 6668,250 kWh and the annual energy sold to the grid is 9909,389 kWh. This table also shows that the amount of electricity purchased varies from 739,831 kWh in December, when the PVM produced the most electricity, to 414,595 kWh in August, when the PVM produced the least.

Figure SM14 depicts the power output throughout the course of the year, revealing that the PVM power production occurred between 06:00 and 18:00, with a peak between 10:00 and 16:00. The output power of the wind turbine is also shown in Figure SM15. According to the graph legend, the maximum experienced wind power for the study area at different times of day and in different months is 12,000 kW. The energy purchased from the grid and sold to the grid throughout the year is represented in Figure SM16. Before 08.00 and after 18.00, most of the energy is sold to the grid.

The total capital cost, total operation and maintenance cost, and total replacement cost of the optimal system were found to be \$20,790,000.00, \$272,862.33 and \$47,371,725.06 respectively. Figure SM17 demonstrates that the PVM accounted for 48.1% of the total capital. The cost of the wind turbine accounted for 38.7% of the total capital cost, followed by the system converter cost, which represented approximately 13.2%.

Sensitivity analysis for the analysis of the optimized configuration of category A

Sensitivity analysis is a tool that provides a good understanding of how target variables are affected based on changes in sensitivity variables. It is a way to predict the outcome of a decision given a certain range of variables. The PVM cost, grid electricity price and inflation rate all have a substantial impact on the LCOE and NPC. The impact of these three sensitivity variables on the system was studied to discover their effect on costs and savings. Although the PVM cost is valued at \$1072,

Table 6

The LCOE and NPC for wide variation in a) PV cost b) grid power price and c) inflation rate.

Parameter	NPC (\$)	LCOE (\$/kWh)
<i>PV cost (\$)</i>		
1072	20.9 Million	0.0118
1572	24.8 Million	0.0139
2072	28.5 Million	0.0160
2572	32.3 Million	0.0181
3072	36 Million	0.0203
<i>Grid power price (\$)</i>		
0.0934	20.9 Million	0.0118
0.1034	25.6 Million	0.0144
0.1134	30.2 Million	0.0170
0.1234	34.8 Million	0.0196
0.1334	39.4 Million	0.0222
<i>Inflation rate (%)</i>		
15.7	20.9 Million	0.0118
16.7	17.9 Million	0.00865
17.7	13.8 Million	0.00576
18.7	8.2 Million	0.00295
19.7	653,256	0.000204

we expect prices to continue to moderately rise owing to exchange rate pressure and technological improvement. In light of this, we established a price range of \$1072 to \$3072, with rates of \$1572, \$2072, \$2572 and \$3072. The grid power price and inflation rate is also varied between \$0.0934 and \$0.01334 and 15.7% and 19.7% respectively. The LCOE and NPC for the variation in PVM cost, grid electricity price and inflation rate for the system economy is displayed in Table 6. The mandated change in the parameters causes a variation in the LCOE and NPC value. We can see from the table that the LCOE is strongly influenced by the initial capital of the PVM. The results show that with the increase in PVM price, the COE and the NPC also increases. The LCOE increases from 0.0118 \$/kWh to 0.0203 \$/kWh when the PVM cost is increased from \$1072 to \$3072. The NPC of the project nearly doubles with a PVM increase of three times the current price. Similarly, the LCOE and NPC increase from 0.0104 \$/kWh to 0.0235 \$/kWh, and from \$20.9 Million to \$39.4 Million, respectively, when the grid electricity price is raised from 0.094 \$/kWh to 0.134 \$/kWh. When the inflation rate is adjusted from 15.7% to 19.7%, the LCOE and NPC changes from 0.0118 \$/kWh and \$20.9 Million to 0.000204\$/kWh and \$653,256, respectively. This shows that the system becomes more feasible in an economy with high inflation. The impact of inflation rate on discounted payback period are illustrated in Figure SM18. It can be seen that with the rise in inflation rate, the discounted payback period decreases. A discounted payback period expresses the number of years required to recoup the initial investment by discounting future cash flows and accounting for the time value of money. So, inflation will benefit investors with large debts because they do so with money that are worth less than the loans they are repaying.

The community will gain from the application of the research findings outlined in this article in the following ways. Increased access to electricity guarantees satisfactory quality of life of the community and comfort, enables students to study after dark, increases safety, and allows use of electrical appliances to simplify house hold task. The availability of electricity will also promote rapid economic development and job creation by fostering the growth of small industries and home-based ventures. It will lessen air pollution and its associated health problems, which will definitely improve public health. A healthier populace will result in a significant reduction in the health budget that individuals and governments should set aside each year. Additionally, it will also lessen the large-scale migration to our metropolises.

Conclusion

This study investigates the feasibility of an on-grid renewable energy solution for electricity generation in a selected location in River State, Nigeria. The investigation was based on techno-economic modelling and optimization and sensitivity analysis. The normal average energy consumption for Choba is estimated to be approximately 39.12–46.8 MW for full 24-hour supply. PHEDC provided 2173 h of electricity in 2021, which is insufficient for a thriving community. In times when there is a fault, frequent blackouts can stretch for hours, sometimes days, leaving homes and businesses without power. The on-grid PVM/wind turbine energy system is the best system configuration for meeting the community load demand at the lowest energy cost along with O&M cost. The configuration showed that this energy system generates annual electricity of 26,605,673 kWh with wind turbine contributing 35.7% and the total net present cost and the levelized cost of energy is found to be \$20.8 Million and \$0.0117/kWh, respectively. It is further noted that the LCOE and NPC are significantly impacted by the price of grid electricity, the PVM cost, and the inflation rate. The results show that implementing this research will assure a consistent power supply to the community while simultaneously feeding the national grid and lowering greenhouse gas emissions into the atmosphere. To advance this study, future research will involve the integration of hydroelectricity as a river runs along the community for power generation in the proposed energy system. Additionally, this study only looked at the technical, environmental, and economic aspect; other perspectives, such as those on energy policy and regulations,

market structure, legal considerations and prerequisites should be taken into account when evaluating energy system sustainability.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.sciaf.2023.e01781](https://doi.org/10.1016/j.sciaf.2023.e01781).

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